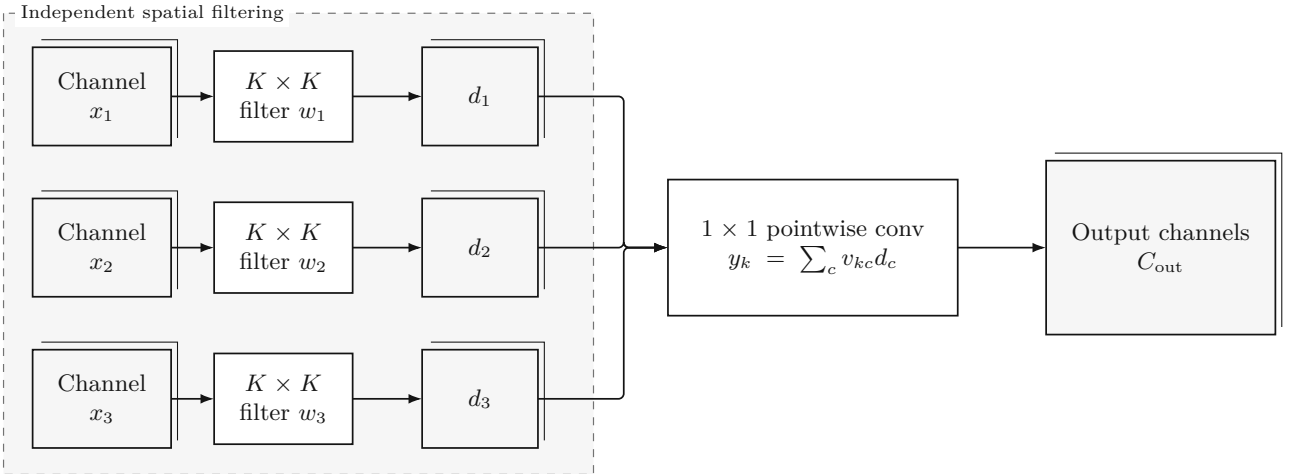
Depthwise separable convolution: split spatial and channel mixing

Conceptual architecture | original schematic | symbolic dimensions



Parameters: $K^2 C_{\text{in}} + C_{\text{in}} C_{\text{out}}$, compared with $K^2 C_{\text{in}} C_{\text{out}}$ for a full convolution.

Depthwise filters act independently on input channels; pointwise filters combine the resulting channels.